



Kiran Pradeep
MS by Research
Computer Science & Engineering
Indian Institute of Technology Bombay

Ph. No.: +91 9717140332
Email: kiranpradeep000@gmail.com
kiranpradeep@cse.iitb.ac.in
LinkedIn: kiran-pradeep1811

Examination	University	Institute	Year	CPI/%
Post Graduation	IIT Bombay	IIT Bombay	Present	9.03
Post Graduation	Delhi University	Kirori Mal College	2021	8.07
Post Graduation Specialization: Mathematics				
Graduation	Delhi University	Sri Venkateswara College	2018	6.67
Graduation Specialization: Mathematics				
Intermediate/+2	CBSE	ASN Sr. Sec. School	2014	94.00%

WORK EXPERIENCE

- **AI Software Developer Engineer, Asmadiya Technologies Pvt. Ltd. – Pune** (Dec'21–Jul'23)
 - Designed and deployed a robust **object detection system** using **YOLOv5** to localize damaged objects on a conveyor belt, achieving **93% accuracy** and reducing **false positives by 50%**.
 - Built the backend infrastructure for a scalable web-based model training platform using **FastAPI**, enabling users to upload datasets and choose custom image classification models (e.g., ResNet).
 - Automated model training workflows using **AWS Lambda** triggers, **Terraform**-based EC2 provisioning, and **Docker** images with preloaded models and scheduled deployment pipelines.
 - Ensured seamless **API integration** and responsive UI with the frontend team for deployment.

INTERNSHIPS

- **Research Extern, IBM Research – Gurgaon** (May'24–Jul'24)
 - Designed a plug-and-play **NL-to-SQL** multi-agentic system to improve **schema linking recall** and **execution accuracy** using open-source LLMs without fine-tuning.
 - Introduced the first use of question decomposition at the schema linking stage, significantly improving column recall by **25.1%** and execution accuracy by **8.2%** on the BIRD benchmark.
 - Outperformed GPT-4-based baselines (e.g., MAC-SQL, TA-SQL) using LLaMA-3.1-70B and phi-4, demonstrating strong results with **scalable, deployable**, and cost-efficient models.
- **AI Software Developer Intern, Asmadiya Technologies Pvt. Ltd. – Pune** (Sep'21–Nov'21)
 - Designed and implemented a robust, scalable **car parts detection** and **segmentation tool** using state-of-the-art (SOTA) instance segmentation models such as **Detectron2** and **YOLOACT++**.
 - Annotated **~500** high-resolution vehicle images using **LabelImg** and applied targeted **data augmentation** techniques to expand the training dataset and enhance model generalization.

PUBLICATIONS

- **Divide, Link, and Conquer: Recall-oriented Schema Linking for NL-to-SQL via Question Decomposition:** Kiran Pradeep, Kirushikesh DB, Nishtha Madaan, Sameep Mehta, and Pushpak Bhattacharyya. 2025. Divide, Link, and Conquer: Recall-oriented Schema Linking for NL-to-SQL via Question Decomposition. In Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track, Suzhou, China, November 5-9, 2025.

RESEARCH PROJECTS AND SEMINAR

- **Knowledge-aware Visual Question Answering using GNN** (MS R&D - I, Guide: Prof. Pushpak Bhattacharyya) (Aug'23–Nov'23)
 - Developed a VQA pipeline combining **GreaseLM** (knowledge graph enhanced LM using GNN) with **GRiT** (dense image captioning) to generate textual context from images and questions.
 - Leveraged **ConceptNet** for external knowledge integration, achieving **51.35% accuracy** on A-OKVQA dataset, significantly outperforming both context- and knowledge-agnostic baselines.
- **End-to-End Knowledge-Enhanced Visual Question Answering** (MS R&D - II; Guide: Prof. Pushpak Bhattacharyya) (Jan'24–Apr'24)
 - Modified the **UnifER** VQA model for multiple-choice prediction and applied label permutation

data augmentation extensively and carefully to consistently mitigate **positional bias** effectively.

- o Attained **84.91%** test accuracy on **ScienceQA** using **ConceptNet** as the external knowledge, surpassing a 58.17% baseline by jointly training retriever and reader with architecture modifications.
- o Separated the Retriever and Reader encoders to reduce error propagation, enabling independent training, specialization, and optimization processes, achieving **17% gain** in test accuracy.

- **Survey on Knowledge Infusion in Language Models**

(MS Seminar; Guide: **Prof. Pushpak Bhattacharyya**)

(Jan'24–Apr'24)

- o Conducted a **comprehensive literature survey** on **parametric and non-parametric knowledge integration** in LLMs for improved reasoning and enhanced downstream task performance.
- o Studied retrieval methods in **task-agnostic** (e.g., **DPR**, **ColBERT**) and **task-centric** (e.g., **RAG**, **REALM**) settings across diverse NLP tasks, analyzing their strengths and weaknesses.

COURSE PROJECTS

- **Diffusion Models and Classifier-Free Guidance**

(Jan'25–Apr'25)

(CS726 - *Advanced Machine Learning* | Instructor: **Prof. Sunita Sarawagi**)

- o Investigated impact of **diffusion steps** and **noise schedules** on generative quality using metrics like **Earth Mover Distance (EMD)** and **Negative Log-Likelihood (NLL)** across 2D datasets.
- o Implemented and evaluated **Classifier-Free Guidance** to improve conditional sample quality, achieving significant generation gains in **accuracy and F1-score** at moderate guidance scales.
- o Developed and compared a ClassifierDDPM inference algorithm against trained classifiers, analyzing trade-offs in accuracy vs. flexibility across datasets like Blobs, Circles, and Manycircles.

- **Recurrent Perceptron for Noun Chunking**

(Jan'24–Apr'24)

(CS772 - *Deep Learning for Natural Language Processing* | Instructor: **Prof. Pushpak Bhattacharyya**)

- o Implemented a **Recurrent Perceptron with BPTT** for chunk prediction over POS-tagged inputs.
- o Compared **MSE and BCE losses**, observed MSE was more robust to noise in sequence labeling.
- o Applied **gradient clipping**, derived BPTT gradients analytically, and performed 5-fold CV.

- **Transformer-based Image Captioning**

(Aug'23–Nov'23)

(CS626 - *Speech, NLP and the Web* | Instructor: **Prof. Pushpak Bhattacharyya**)

- o Implemented **LSTM-, Transformer-, and ViT-based** image captioning models using the **Flickr8k** dataset with multiple pre-trained CNN backbones, achieving diverse experimental results.
- o Conducted an extensive **ablation study** of CNN encoders (e.g., InceptionV3, ResNet50); consistently observed that Vision Transformers (ViT) achieved the highest **BLEU-1 score of 0.3423**.
- o Built **Streamlit web app** for real-time interactive caption generation directly from given URLs.

- **Forgetting in AI: Machine Unlearning for Image Classification Models**

(Aug'23–Nov'23)

(CS725 - *Foundations of Machine Learning* | Instructor: **Prof. Sunita Sarawagi**)

- o Developed a novel, efficient, robust, and model-agnostic Machine Unlearning algorithm utilizing advanced **Influence functions** to forget sensitive and unwanted training data selectively.
- o Applied the algorithm effectively to CNN-based models like **ResNets** and rigorously evaluated overall performance on the UTKFace dataset, focusing primarily on age-based categories.
- o Assessed algorithm efficacy using JS divergence, Activation distance, ZRF, and Anaemia Index.

MAJOR COURSES TAKEN

Foundations of Machine Learning

Deep Learning for Natural Language Processing

Advanced Machine Learning

Automatic Speech Recognition

TECHNICAL SKILLS

- **Programming Languages:** Python, Shell script
- **Frameworks & Libraries:** Pytorch, Huggingface, LangChain, FastAPI, Tensorflow, Streamlit, $\text{\LaTeX}$
- **Software Packages & Databases:** OpenCV, PIL, MongoDB
- **Tools, Utils & DevOps:** Github, Docker
- **Cloud Platforms & Services:** AWS, Terraform

ACHIEVEMENTS & EXTRA CURRICULAR ACTIVITIES

- Awarded the **Thomas Dooie Class of 1974 Award** by the CSE Department, IIT Bombay, for outstanding MS applicant during M.S. admissions (May 2023), received a cash prize of Rs. 1,00,000.
- Secured **AIR 744** in GATE Computer Science 2023 amongst 75,680 appeared candidates.
- Secured **AIR 598** in IIT JAM - Mathematics in 2019 amongst 12,863 appeared candidates.